

Project Design Phase

Problem Solution Fit & Testing Validation

Date	30 July 2026	Team ID	Individual / Solo Project
Student	Bandhanadam Antony	Project Name	Automated Network Request Management in ServiceNow

The Problem-Solution Fit ensures that the discovered customer problem is accurately addressed by the realized solution. It helps identify behavioral patterns, constraints, and the most effective automation channels. For the Network Request Management project, it maps the transition from manual, email-based firewall/port requests to a fully automated IntegrationHub deployment.

1. Customer Segment(s) • Employees/Developers needing network, VPN, or port access • Managers who approve access requests • Network & InfoSec Admins who audit and deploy configurations	2. Jobs-to-be-Done / Problems • Employee: Get firewall ports opened or VPN access granted rapidly. • Manager/InfoSec: Review requests against security compliance efficiently. • IT: Deploy changes accurately to endpoints without manual typing.	3. Triggers • Deploying a new application requiring cross-server communication. • Onboarding a remote contractor needing VPN. • Troubleshooting a blocked system integration.	4. Emotions: Before / After • Before: Blocked, frustrated, uncertain of IT SLA. • After: Productive, secure, and empowered by fast self-service.	5. Available Solutions • Manual email requests to Network Operations • Informal chat messages to IT staff • Cons: Highly error-prone, zero security audit trail, slow manual CLI configuration.
6. Customer Constraints • No dedicated self-service forms (reliance on generic IT tickets) • Lack of visibility into InfoSec policy constraints • Network topologies tracked manually, causing configuration delays	7. Behaviour • Employees guess IP addresses and ping admins directly. • InfoSec manually reviews spreadsheets to check active firewall rules. • Admins manually SSH into routers to apply rules.	8. Channels of Behaviour • Online: ServiceNow Service Portal, Now Mobile App, Automated Email/Teams alerts. • Offline: IT Helpdesk walk-ins, daily stand-up meetings.	9. Problem Root Cause • No standardized, digitized workflow linking the user's request, the security policy validation, and the actual network device orchestration.	10. Your Solution • Service Catalog: Standardized self-service portal items with CMDB data lookups. • Flow Designer: Automated, multi-tiered Manager & InfoSec approval routing. • IntegrationHub & Ansible: Zero-touch automated deployment directly to network devices. • Real-time Status: Automated

Testing & Validation Phase

Objective: Validate end-to-end request lifecycle and automation.

- Perform unit testing on each Service Catalog item (VPN, Firewall Rule, Port Access) to ensure UI policies and client scripts function correctly.
- Conduct end-to-end request simulation with multiple scenarios (Standard vs. High-Risk requests).
- Validate manager and InfoSec approvals, automated fulfillment task creation, email notifications, and CMDB data accuracy.
- Involve key business stakeholders and Network Admins in UAT (User Acceptance Testing) to ensure the solution meets operational needs.
- Log defects via the ServiceNow Agile/Defect application and iteratively deploy improvements.